CartPole DQN Extension - One-Page Summary

Adithya V

Context

GridWorld (500 seeds, tabular Q) approved. Professor instructed extension to CartPole-v1 + Double DQN (main_cartpole_dqn.py). Results: cartpole_experiments_fair_v2.csv (5 seeds).

GridWorld (approved baseline, n=500)

	Clean	Under attack*
Vanilla	7.35 +/- 0.24	-682 +/- 329
Pessimistic	7.35 +/- 0.27	-22 +/- 50
Off-policy	7.37 +/- 0.23	-552 +/- 335

*Vanilla vs pes attacker; pes/off vs own. Fixed attackers. Pes ~11x slower under attack.

CartPole fair eval (extension, n=5)

Metric	Vanilla	Pessimistic	Off-policy
Clean	421 +/- 176	485 +/- 33	485 +/- 20
Matched attacker	239 +/- 247	246 +/- 238	153 +/- 136
Shared random	344 +/- 224	329 +/- 197	365 +/- 206
Worst-case (32-d)*	232 +/- 182	280 +/- 141	243 +/- 124

*Defender-specific worst case. Pes inference ~2.0 ms/step vs greedy ~0.04 ms (~52x).

Per-seed clean (V/P/O): 0:500/500/500 1:500/500/500 2:107/500/500 3:500/500/464 4:500/426/461

Takeaways (descriptive only)

1. Extension runs end-to-end; pessimistic stable after v2 training fixes.
2. No single ordering across CartPole metrics (unlike GridWorld).
3. Pes highest mean worst-case; off highest shared-random; off lowest matched.
4. Robustness-speed tradeoff persists (maximin vs greedy inference).
5. Not full BP-DQN; simplified belief; n=5.

Files & reproduce

File	Role
cartpole_experiments_fair_v2.csv	Results (dedupe by seed)
main_cartpole_dqn.txt -> .py	Code
cartpole_belief.py	Belief RNN
gridworld_experiments_500runs.csv	Approved baseline

```
pip install -r requirements.txt
cp main_cartpole_dqn.txt main_cartpole_dqn.py
python3 main_cartpole_dqn.py --step 8 --csv cartpole_experiments_fair_v2.csv
```